When Better Gets Worse: Improvement Fidelity for Self-Improving Agents in Adaptive Worlds

Anonymous Authors

Abstract

Self-improving agents repeatedly replace an incumbent policy with a candidate that appears better under a verifier or world model. Existing evaluation usually measures value or ranking of isolated policies. We argue that the statistical object that must transfer is the update itself, $\pi \to \pi'$. We define *improvement fidelity*, the agreement between proxy and deployment-induced improvement, and call $\Delta_V > 0$ with $\Delta_* < 0$ *improvement reversal*. We introduce PIVOT (Paired Interventional Validation of Optimization Transitions), which models paired differential corrections and allocates expensive intervention queries to updates likely to change the selection decision. Two elementary propositions show that global value accuracy is sufficient but stronger than needed, while update error is controlled by response sensitivity times update footprint under a Lipschitz response assumption. In a controlled performative environment, three registered runs reproduce high-response reversal (IRR 0.9722) and PIVOT reduces one-query selection regret relative to random and top-proxy querying. Adaptive opponents produce strategic reversal in the fixture (SIRR 1.0). An observational Binance futures audit across 12 asset/date sessions finds a negative depth-based mechanical effect but no reversal (0/7 primary, 0/5 holdout), so it is not treated as causal evidence. The result is a scoped evaluation framework and a reproducible boundary between controlled mechanism evidence and external validity, rather than a claim that every learned world or market simulator is faithful.

1 Introduction

Self-improvement is often described as a loop: generate a candidate, evaluate it, promote it, and repeat. The loop hides a subtle change of estimand. A verifier may correctly evaluate the world it implements while omitting the consequences caused by deploying the candidate. A policy can therefore improve on the verifier's world and deteriorate in the world its deployment induces. This failure is distinct from self-authored verification, ordinary dataset overfitting, or an incorrect evaluation program.

We study the local transition

$$\tau_t = (\pi_t, \pi_{t+1}) \qquad \text{rather than only} \qquad J(\pi_t). \tag{1}$$

Let V be a cheap observer or replay world and let $*$ denote a deployment world that may respond to the policy. The proxy and deployment improvements are $\Delta_V = J_V(\pi') - J_V(\pi)$ and $\Delta_* = \mathcal{J}(\pi') - \mathcal{J}(\pi)$. An improvement reversal is

$$\Delta_V > 0 \quad \wedge \quad \Delta_* < 0. \tag{2}$$

This definition makes no assumption about whether the proxy is learned, hand-coded, or externally audited. It asks whether the object consumed by the self-improvement loop transfers.

Our contributions are deliberately limited to three points:

1. We formalize **improvement fidelity** and report differential metrics for local update decisions: improvement differential error (IDE), sign consistency (ISC), reversal rate (IRR), strategic reversal rate (SIRR), update-selection regret (ISR), and cumulative true improvement (CTI).
2. We introduce **PIVOT**, a paired correction model plus active high-fidelity querying rule. It spends intervention budget where a query is most likely to change the update ordering or sign.
3. We provide a controlled performative evaluation, a strategic fixture, and a checksum-bound observational finance audit. We preserve nulls and explicitly separate fixture evidence from causal and external claims.

2 Related Work

Performative prediction formalizes the feedback from a deployed predictor to the data-generating process [7]. Performative RL and recent policy-gradient analyses extend this idea to policy-dependent transition and reward dynamics [1]; our focus is the local validity of a proposed update rather than optimization of a performatively stable policy. WorldGym and policy-aware simulator learning study whether policy values and rankings transfer from a learned world to a target environment [8, 4]. We instead treat the paired difference between an incumbent and a candidate as the estimand.

Self-evolving strategy systems and policy-evolution benchmarks make iterative improvement executable [6, 10], while self-authored verification work exposes failures when an agent controls its own tests [5]. PIVOT addresses a distinct failure mode: even a fixed external verifier can approve an update that is harmful after deployment-induced response. Multi-agent performative prediction and endogenous market simulators motivate the strategic layer [9, 3]; they are testbeds here, not the paper's primary novelty.

3 Problem Formulation

Policy-induced worlds. Deployment can alter transitions, rewards, liquidity, or other agents' behavior. We write this response as $\mathcal{M}[\pi]$ and define the deployment value

$$\mathcal{J}(\pi) = J(\pi; \mathcal{M}[\pi]). \tag{3}$$

For exposition, three layers separate mechanisms rather than assert that they are always identifiable:

$$\Delta_{\text{direct}} = J(\pi'; \mathcal{M}[\pi]) - J(\pi; \mathcal{M}[\pi]), \tag{4}$$
$$\Delta_{\text{actor}} = J(\pi'; \mathcal{M}[\pi']) - J(\pi; \mathcal{M}[\pi]), \tag{5}$$
$$\Delta_{\text{strategic}} = J_i(\pi', BR_{-i}(\pi')) - J_i(\pi, BR_{-i}(\pi)). \tag{6}$$

The first ignores response, the second includes the focal agent's mechanical footprint, and the third includes adaptation by competitors. In experiments we report these as separate worlds and do not label a learned or observational proxy as ground truth.

Improvement fidelity metrics. For n transitions, IDE is $n^{-1} \sum_k |(\Delta_V)_k - (\Delta_*)_k|$. ISC is the fraction of non-tied transitions whose signs agree. IRR is $P(\Delta_* < 0 \mid \Delta_V > 0)$, while SIRR is $P(\Delta_{\text{strategic}} < 0 \mid \Delta_{\text{actor}} > 0)$. For a candidate set C_t, the update-selection regret is

$$\text{ISR}_t = \max_{j \in C_t} (\Delta_*)_{t,j} - (\Delta_*)_{t,\hat{j}}, \tag{7}$$

where $\hat{j}$ is selected by the proxy or corrected estimate. CTI sums the true improvements of promoted updates over a trajectory. All estimates below retain query counts, seeds, and confidence intervals in the artifact snapshot.

4 PIVOT

PIVOT stands for Paired Interventional Validation of Optimization Transitions. Given an incumbent and K candidates, it first computes cheap proxy deltas and pre-query features. The features include proxy delta, response strength, competition strength, candidate index, and an update footprint. The generic footprint records policy-distance and action/support shifts; the finance footprint records size, participation, urgency, holding, turnover, liquidity consumption, and spread crossing.

Paired differential correction. For a queried transition, incumbent and candidate are evaluated on the same initial state, exogenous path, seed, and opponent initialization. PIVOT fits

$$g_\theta(z_\Delta) \;\approx\; \Delta_* - \Delta_V, \qquad \widehat{\Delta}_* = \Delta_V + g_\theta(z_\Delta). \tag{8}$$

Pairing subtracts common-mode noise before aggregation. The unpaired ablation uses disjoint context seeds and the standard independent-sample standard error.

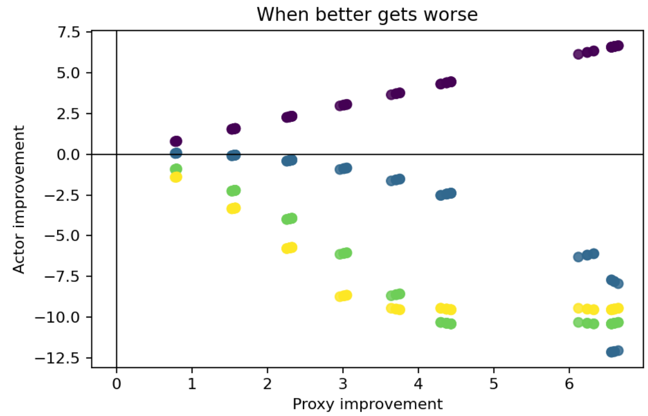

Figure 1: Controlled proxy versus deployment improvement. The lower-right quadrant is improvement reversal. Each point is a transition in the frozen response–footprint grid.

Active query rule. Let $\widehat{\Delta}_j$ and u_j be a corrected estimate and uncertainty. A transparent PIVOT score is

$$q_j = \frac{u_j(0.5 + r_j)(1 + o_j)}{c_j}, \tag{9}$$

where r_j is predicted sign risk, o_j is opportunity relative to the current best candidate, and c_j is high-fidelity cost. The rule is a decision-change heuristic, not a Bayesian optimality claim. Queried values replace predictions; unqueried candidates retain proxy/model estimates.

5 Theory

We state two modest results that motivate the differential estimand. They are not claims about a particular simulator architecture.

Proposition 1 (Global fidelity is sufficient)**.** *Suppose* $\sup_\pi |J_V(\pi) - J_*(\pi)| \le \varepsilon$. *Then every transition satisfies* $|\Delta_V - \Delta_*| \le 2\varepsilon$. *In particular, the sign is preserved whenever* $|\Delta_*| > 2\varepsilon$.

Proof. By adding and subtracting the two policy values, $\Delta_V - \Delta_* = [J_V(\pi') - J_*(\pi')] - [J_V(\pi) - J_*(\pi)]$. The triangle inequality gives the bound. If $|\Delta_*| > 2\varepsilon$, the perturbation cannot cross zero. □

Proposition 2 (Response–footprint bound)**.** *Let* $d(\pi, \pi')$ *be an update footprint and define the direct comparison in the incumbent response world,* $\Delta_{\text{direct}} = J(\pi'; \mathcal{M}[\pi]) - J(\pi; \mathcal{M}[\pi])$. *Assume the environment response satisfies* $D(\mathcal{M}[\pi'], \mathcal{M}[\pi]) \le L_M d(\pi, \pi')$ *and the value functional is* L_J*-Lipschitz in its world argument. Then*

$$|\Delta_{\text{actor}} - \Delta_{\text{direct}}| \le L_J L_M d(\pi, \pi'). \tag{10}$$

Proof. The incumbent term cancels because both comparisons use $\mathcal{M}[\pi]$ as the reference world. The remaining candidate term is bounded by $L_J D(\mathcal{M}[\pi'], \mathcal{M}[\pi]) \le$

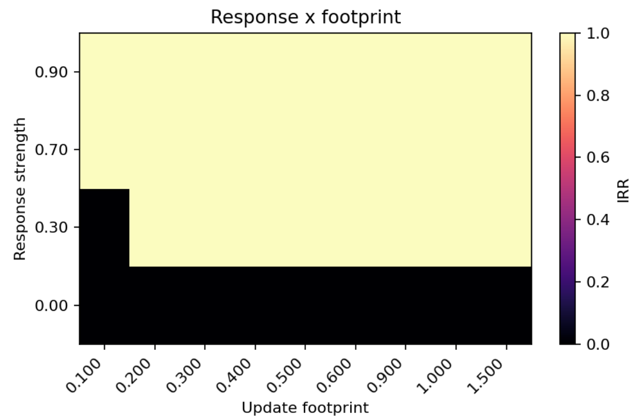

Figure 2: Reversal rate over response strength and update footprint. The structure is a controlled mechanism result, not a universal market law.

$L_J L_M d(\pi, \pi')$. The result is local; it does not assert that strategic best responses are globally Lipschitz. □

The first proposition explains why global value accuracy is sufficient but unnecessarily strong: a policy-independent offset can be arbitrarily large without changing any improvement difference. The second motivates targeted fidelity: high footprint and high response sensitivity are the transitions most likely to change a decision. In strategic environments, small policy changes can still cause large competitor responses, so the bound should be treated as a diagnostic hypothesis rather than a universal smoothness assumption.

6 Controlled Experiments

Setup. The controlled world has finite horizon 12, bounded rewards, Gaussian exogenous noise, and a tunable response strength. Candidate policies change one intensity component. P2 uses three independent registered run groups and a response grid $\{0, 0.3, 0.7, 0.9\}$, footprint scales $\{0.1, 0.2, 0.3, 0.5\}$, and optimization strengths $\{0.5, 1, 1.5\}$. All primary comparisons use disjoint registered seeds and paired high-fidelity transitions.

Improvement reversal. Across 240 transitions per registered run, overall IRR is 0.7292 and the high-response IRR is 0.9722; low-response IRR is zero. The phase diagram in Figure 2 shows the intended response–footprint interaction. These results establish that a correct observer verifier can approve harmful updates in a world with policy-induced response.

Global versus differential fidelity. The matched-budget E4 comparison trains a global value model and a transition differential model on the same number of high-fidelity transition queries. Local-minus-global ISC is 0.1204; local-minus-global IDE is -1.7536. Update-selection regret is tied in this particular fixture, so we

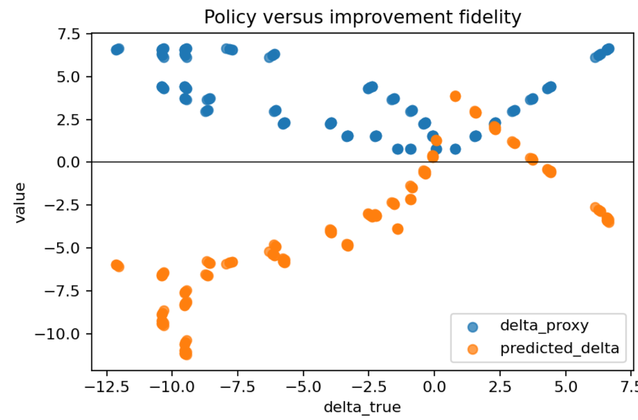

Figure 3: Policy-value and transition-differential predictions under matched high-fidelity budget. The estimands are related but not interchangeable.

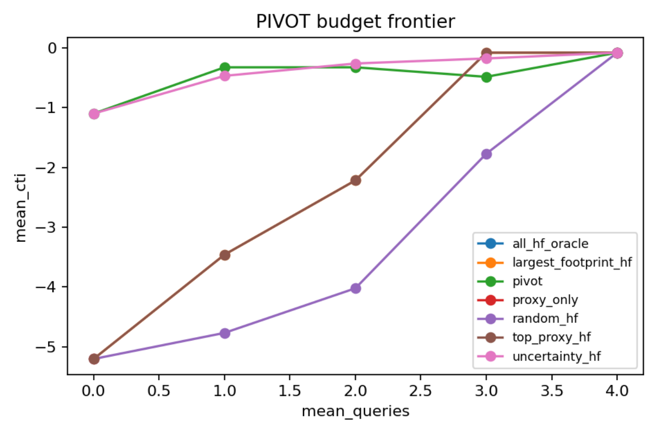

Figure 4: Matched-budget selection frontier. PIVOT spends queries on decision-sensitive transitions; All-HF is an oracle reference.

do not claim that a differential model dominates every policy-value baseline.

Budget frontier. At one high-fidelity query, PIVOT reduces ISR relative to Random by 4.7245 and relative to Top Proxy by 3.1301 in the registered fixture. Figure 4 shows the full frontier. The twelve-ablation suite retains a non-monotonic PIVOT curve across budgets; this is a robustness boundary, not a claim of monotonic benefit.

A required null. The E3 closed-loop fixture does not show proxy/true divergence: both curves increase together and reach the reward bound. We therefore report E3 as a negative stress test of the current fixture, not as evidence for "optimizing the wrong world." This distinction is important because a paper can otherwise manufacture its central phenomenon by selecting a pathological environment after observing results.

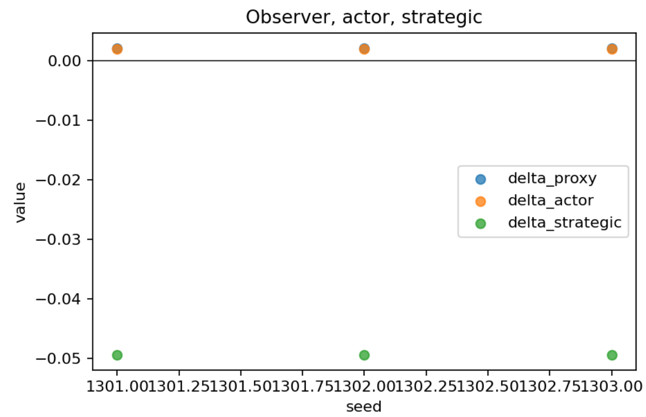

Figure 5: The same update evaluated in observer, actor, and strategic worlds.

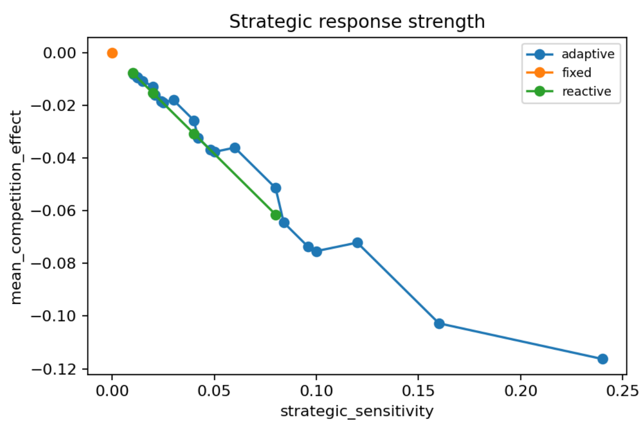

Figure 6: Strategic response effect across opponent modes and sensitivities.

7 Strategic and Finance Testbeds

Strategic response. Only the focal policy self-improves. Competitors are fixed, reactive, or finite- step adaptive. In the registered fixture, actor improvement is positive before adaptation, while adaptive competition produces SIRR 1.0 and competition effects of -0.0514 (E7) and -0.0405 (E8). The sensitivity contrast is -0.1082.

These results are mechanism evidence for the strategic layer, not a claim of general equilibrium or realistic market ecology. External strategic validation remains open.

Finance audit. The finance ladder is F0 historical path, F1 execution replay, F2 interactive virtual fills, and F4 strategic response. We additionally audit 12 sessions covering BTCUSDT, ETHUSDT, and BNBUSDT at four 2023 quarter starts using Binance's public one-minute klines and percentage book depth [2]. The archive checksums are frozen in the project manifests. The depth world is an observational execution proxy: it cannot identify replenishment, hidden liquidity, post-trade response, or strategic adaptation.

At primary participation, seven sessions have positive F1 improvement and depth-proxy reversal is 0/7; the frozen holdout has 0/5 reversals. The pooled depth mechanical effect is -4.1554×10^{-7} with 95% bootstrap interval $[-5.3969, -2.9638] \times 10^{-7}$. Thus finance supplies a plausibility and boundary test, not causal confirmation of improvement reversal.

Table 1: Selected controlled ablations. Values are descriptive estimates over three independent registered runs; slash-separated values compare variants.

Ablation	Estimate(s)	Metric
Paired vs unpaired	0.0433 / 0.0676	paired SE / unpaired SE
Footprint vs none	2.0668 / 2.0669	IDE; null
Small vs large	0.50 / 0.75	IRR
Weak vs strong response	0.458 / 1.000	IRR
Fixed vs adaptive	0.000 / 1.000	SIRR
Single vs multiple model	4.751 / 2.691	IDE
Candidate count 2 / 4	0.2368 / 0.2439	ISR
PIVOT budget 0 / 4	1.0210 / 0.0000	ISR

8 Ablations and Robustness

The frozen suite contains twelve required comparisons. Table 1 shows the compact main-text subset; the full aggregate, raw rows, and failure ledgers are included in the artifact snapshot.

Pairing reduces standard error (0.0433 versus 0.0676). Multiple response levels improve held-out differential error relative to a single response level. However, footprint features are nearly tied with no-footprint features in this grid (IDE 2.0668 versus 2.0669), and candidate-count/HF-budget effects are small or non-monotonic. These nulls constrain the method claim: PIVOT is a budgeted transition-validation mechanism, not proof that every feature or budget increase helps.

9 Limitations and Claim Boundary

First, the controlled world is intentionally small and its response law is known. It establishes falsifiability of the estimand, not realism. Second, the public finance data are observational and use virtual fills; no live orders, broker connection, or private L2 data were used. Third, the strategic fixture is deterministic and requires validation in an independently calibrated competitive environment. Fourth, the current public typed update was selected exploratorily; a confirmatory update-generation and regime/holdout rule must be frozen before a new external claim. Fifth, E3's null means the current fixture does not demonstrate performative overoptimization. Finally, LLM/EvoQuant and M3/world-model adapters are intentionally deferred and are not needed for the core estimand.

The claim boundary is therefore precise: we validate learning progress at the policy-transition level in controlled adaptive worlds, demonstrate a strategic mechanism in a fixture, and report a negative observational finance audit. We do not certify actions, solve general multi-agent equilibrium, or claim that a learned simulator is ground truth.

10 Reproducibility and Conclusion

The paper snapshot contains 25 hash-indexed artifacts, seven figure/source-table pairs, controlled gate summaries, the public expansion summary, and the full twelve-ablation aggregate. Three disjoint ablation runs reproduce with zero failed runs. The repository's full test suite reports 82 passing tests; Ruff and mypy are clean. The build records the source commit and PDF verification metadata. No credentials or raw vendor archives enter the paper package.

The central lesson is simple: a self-improving agent should not be judged only by whether its new policy performs better in the world used to verify it. It should be judged by whether the update remains better in the world—and among the agents—that deployment causes to exist. PIVOT operationalizes this as paired, transition-level, budgeted interventional validation. The controlled evidence supports the need for that estimand; the external audits state exactly where evidence is still missing.

References

[1] Debabrota Basu, Udvas Das, Brahim Driss, and Uddalak Mukherjee. Performative policy gradient: Optimality in performative reinforcement learning, 2025.

[2] Binance. Binance public data repository. `https://github.com/binance/binance-public-data`, 2023.

[3] Patrick Cheridito, Jean-Loup Dupret, and Zhexin Wu. Abides-marl: A multi-agent reinforcement learning environment for endogenous price formation and execution in a limit order book, 2025.

[4] Christoph Dann, Yishay Mansour, and Mehryar Mohri. Theoretical foundations and effective algorithms for policy-aware simulator learning, 2026.

[5] Diandian Guo, Cong Cao, Fangfang Yuan, Yingqi Wang, Yueshan Wang, and Dakui Wang. Self-authored verification is unreliable in heuristic self-improving agents, 2026.

[6] Jie Mao, Changlun Li, Xiang Li, Qiqi Duan, Jinhui Yuan, Xiang Liu, Yuyu Luo, Jing Tang, Xiaowen Chu, and Nan Tang. Evoquant: Self-evolving verifier-guided strategy optimization for robust quantitative trading, 2026.

[7] Juan C. Perdomo, Tijana Zrnic, Celestine Mendler-Dunner, and Moritz Hardt. Performative prediction. *arXiv preprint arXiv:2002.06673*, 2020.

[8] Julian Quevedo, Ansh Kumar Sharma, Yixiang Sun, Varad Suryavanshi, Percy Liang, and Sherry Yang. Worldgym: World model as an environment for policy evaluation, 2025.

[9] Guanghui Wang, Krishna Acharya, Lokranjan Lakshmikanthan, Juba Ziani, and Vidya Muthukumar. Last-iterate convergence for symmetric, general-sum, 2×2 games under the exponential weights dynamic, 2025. Includes a multi-agent performative prediction application.

[10] Zhilin Wang, Han Song, Runzhe Zhan, Jusen Du, Jiacheng Chen, Tianle Li, Qingyu Yin, Yulun Wu, Zhennan Shen, Tong Zhu, Yanshu Li, Guanjie Chen, Derek F. Wong, Yafu Li, Yu Cheng, and Yang Yang. Evopolicygym: Evaluating autonomous policy evolution in interactive environments, 2026.

A Artifact and Full Result Notes

The frozen artifact manifest is `paper/snapshot/manifest.json`. Its controlled summary hashes are inherited from the clean-room reproduction. The ablation aggregate is identified by the manifest hash prefix `e081f87f`; the public expansion remains marked `causal_impact_identified=false` and `live_orders=false`. Re-running the snapshot and table scripts from a new output directory is required before changing any result table.

Table 2: Controlled and public evidence summary.

Experiment	Estimate	Interpretation
P2 reversal	0.9722	IRR, high response
E4 local–global	0.1204	ISC difference
E5 Random–PIVOT	4.7245	ISR at HF budget 1
E5 Top Proxy–PIVOT	3.1301	ISR at HF budget 1
E6 zero participation	0.0000	F2–F1
E7 SIRR	1.0000	strategic reversal
E8 SIRR	1.0000	strategic reversal
E9 CTI	-1.5338	8-round exploratory loop

Table 3: Public expansion boundary.

Public audit field	Value
Asset/date sessions	12
F1-positive sessions	7
Primary reversals	0/7
Holdout reversals	0/5
Pooled depth effect	-4.1554×10^{-7}
Causal impact identified	No

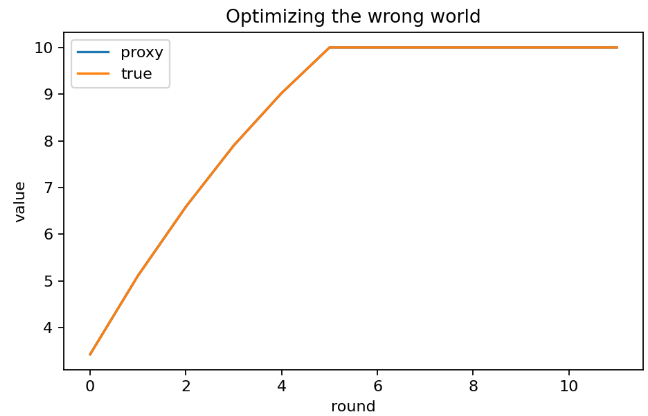

Figure 7: Required E3 null: proxy and true values overlap and saturate at the reward bound. The current closed-loop fixture therefore does not establish performative overoptimization.